Programme 2: Family Patterns

1. What are the sex cells that the body produces called? Gametes
2. What do the cells form in females? Eggs Males? Sperm
3. What is this particular secondary cell division of the sex cells called? Meiosis
4. How many chromosomes do most human body cells contain? 46, or 23 sets
5. How many chromosomes do the gametes of each sex cell contain? 23
6. Why is fertilisation necessary? To reproduce. The moment of conception, when the egg is **fertilised** by the sperm, is the only time in our life cycle when two cells combine to form one cell with equal number of chromosomes from each parent.
7. If chromosomes combine from the same two parents, why do children in a family show all kinds of genetic differences? And why are some male and some female? Its all in your hand-me-down genes ... different eggs and different sperm must carry different genetic information. X **and Y chromosomes** are split during meiosis and using a punnet square we can determine if the offspring are left with XX or XY.
8. Name at least one recessive (Cystic Fibrosis and Down Syndrome) and one dominant (Huntington's and Achonroplasia) condition.
9. State whether the following processes belong to mitosis or meiosis:

Occurs in body cells	**Chromosomes in cell duplicate** _____	**Identical chromosome sets separate.** _____
Cell contents duplicate _____	**Cell divides to form 2 cells of equal size.** _____	**This process happens in body cell of plants & animals.** _____
Occurs in sex cells _____	**Chromosomes duplicate** _____	**Form two identical pairs of chromotids** _____
Centromere is their joining point _____	**4 copies of each chromosome in the cell** _____	**Nucleus membrane disappears** _____
Spindle fibres grow across the cell _____	**Chromosome pairs line up with spindle fibres attached to centromeres** _____	**Spindle fibres contract, separating members of each pair** _____
Each chromosome still has 2 chromatids _____	**Division occurs again. Each cell now has half chromosomes of original cell** _____	**These cells are either sperm or egg** _____